Salesforce Summer Program – Day 5

Goal for Today

By end of today, you should understand:

- What is Apex?
- Why Salesforce needs programming in addition to clicks/flows
- Basic Apex syntax and logic
- Difference between declarative vs programmatic development
- How Apex connects to business logic

Trailhead Modules (COMPLETE TODAY)

Follow this order strictly:

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1 Apex & .NET Basics (~1 hr 10 mins)

Focus on:

- What is Apex
- Variables
- Classes
- Conditional logic
- Loops

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2 Apex Basics & Database (~2 hrs 45 mins)

Focus on:

- Apex syntax
- DML operations
- SOQL basics
- Data manipulation

Do NOT rush. Understanding is more important than completion.

□ Videos (Verified Working Links)

1️⃣ What is Apex in Salesforce? (BEST beginner intro)

https://www.youtube.com/watch?v=F0PHdBzoIdU

Focus:

- Why Apex exists
- How Apex works
- Hello World example

2️⃣ Salesforce Apex Tutorial for Beginners

https://www.youtube.com/watch?v=yuUmKre-Peo

Focus:

- Variables
- Loops
- Conditions
- Classes
- Triggers overview

3️⃣ Apex Programming Crash Course

https://www.youtube.com/watch?v=HeRZjnuXJUQ

Focus:

- Collections
- Logic building
- Exception handling
- Salesforce CLI basics

□ CORE TASKS (MANDATORY)

Task 1 — Connect Everything Learned Till Now

Take your College Management System.

Now integrate:

Concept	Example
CRM	Student admission pipeline
Objects	Student, Course, Faculty
Relationships	Student ↔ Course
Validation	Email required
Formula	Remaining seats
Flow	Auto notification
Apex	Custom business rule

Task 2 — Apex Thinking Exercise

Suggest 3 cases where:
Flow is NOT enough
Custom Apex code is needed

Examples:

- Complex fee calculation
- Integration with external payment system
- Advanced eligibility logic

Explain WHY.

Task 3 — Simple Programming Logic

Write pseudocode (NOT actual Apex necessarily):

Example:
IF seats are full
THEN block registration

OR

IF attendance < 75%
THEN notify student

Task 4 — Reflection Task (MOST IMPORTANT)

Answer:

 Why can’t all enterprise logic be built using only clicks and configuration?

GitHub Submission

Create:

/day5-apex-introduction/

README.md must include:

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1. What is Apex?

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2. Difference:

- Flow vs Apex
- Configuration vs Coding

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3. Real Examples Where Apex Is Needed

(3 examples)

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4. Integrated System Design

Explain your complete College Management System using:

- CRM
- Objects
- Relationships
- Validation
- Flow
- Apex

5. Pseudocode Examples

(Add your logic examples)

6. Reflection

Why enterprise systems eventually need programming

✍️ Reflective Questions

1. Why is Apex needed if Salesforce already has Flows?
2. When should developers prefer no-code solutions?
3. What problems require custom programming?
4. Why is business logic important in enterprise systems?
5. Why should developers avoid unnecessary coding?
6. How does programming increase flexibility?

Rules

- Focus on understanding concepts deeply
- Do NOT memorize syntax blindly
- Think in terms of real business problems
- Spend around 3–4 hours maximum

End of Day Outcome

You should now clearly understand:

- Why Apex exists
- How programming fits into Salesforce
- Difference between declarative and programmatic logic
- How all concepts learned so far connect together